Gauge pinning, a rational boundary Schur gap and nonlinear Haar response on a finite cell

Project Stardust

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Abstract

We reconstruct a finite plaquette-incidence problem with 48 vertices, 112 edges and 72 faces. The 32 interior edges have a one-dimensional gauge kernel; pinning one edge gives an invertible 31-coordinate interior Gram matrix without changing its curvature range. Eliminating those interior variables produces a boundary Schur form whose kernel is exactly the 46-dimensional boundary-gradient space. On its 34-dimensional orthogonal complement we prove a computer-assisted rational lower bound 0.69639 and give an exact nonzero Rayleigh witness strictly below 0.6964. Complete geometric and linear-algebra arguments identify the physical norm, cycle compression and residual certificate. The finite arithmetic is supplied as explicit integer tables with an independent standard-library checker. A separate periodic coarsening has incompatible raw gauge averaging and distinct physical norm/energy minimizers, with exact costs and a feasible step reaching the energy minimum. A periodic gauge counterexample and a perturbation cross-term test delimit which invertibility and stability conclusions transfer. An actual sparse Wilson Hessian has a linear error despite quadratic loss in its scalar Schur complement. We then reconstruct exact nonlinear tests: a two-axis Haar parity, a three-axis full 32-link variance difference, and a separate conditional large-coupling obstruction to a quadratic boundary budget. A valid linear budget remains. Exact nonlinear Haar pinning, a global link-defect bound and a normalized full-cell concentration theorem connect the same pin to the actual positive Hessian at identity. These finite results establish no multiscale or continuum mass gap; Schur elimination and the Haar integration mechanisms are classical.

1 The actual finite problem

Let e_0, e_1, e_2, e_3 be the coordinate unit vectors in $\mathbb{R}^4$. Coordinates and directions in this paper are zero based, matching the formal source. The base vertices are $V_0 = \{0,1\}^4$. The interior oriented edges are

$$E_I = \{(v,i) : v \in V_0,\ v_i = 0\},$$

with source v and target $v + e_i$. There are $4 \cdot 2^3 = 32$ such edges. For each $v \in V_0$ with $v_j = 1$, add a cap vertex $v + e_j$ and an outward edge from v to $v + e_j$. These cap vertices are distinct because they have exactly one coordinate equal to two. There are 32 caps and outward edges.

For every interior edge (v,i) and every $j \neq i$ with $v_j = 1$, add a tangential cap edge from $v + e_j$ to $v + e_i + e_j$. These give 48 tangential edges: choose the ordered distinct directions (i,j) in 12 ways and the remaining two binary coordinates in four ways. Let E_B consist of the 32 outward and 48 tangential edges. There are 80 boundary edges, 48 total vertices V, and 112 total edges $E = E_I \sqcup E_B$.

There are two sorts of faces. The 24 inner squares have base $v \in V_0$, directions $i < j$ and $v_i = v_j = 0$. The 48 ribbons correspond to the triples (v,i,j) used for tangential edges. A ribbon

traverses the interior edge (v, i), the outward edge at $v + e_i$ in direction j, the negative tangential cap edge and the negative outward edge at v in direction j. Thus each is a genuine oriented square. Put $F = F_I \sqcup F_R$, of size 72.

Object	Construction	Count
Base/cap vertices	$16 + 32$	48
Interior edges	$4 \cdot 8$	32
Outward/tangential boundary edges	$32 + 48$	80
Inner squares/ribbons	$24 + 48$	72

Source correspondence: `OneSidedBlockIncidence`
`edges_card; innerFaces_card; upperFaces_card`

Source correspondence: `OneSidedCapComplex`
`boundary_card; fullVertex_card; fullEdge_card; fullFace_card`

All edge and face spaces have their ordinary Euclidean norm, with every edge and face counted once. For a potential $\phi \in \mathbb{R}^V$, define $d_0\phi(e) = \phi(\text{target}\, e) - \phi(\text{source}\, e)$. For $f \in \mathbb{R}^E$, let $d_1 f \in \mathbb{R}^F$ be the signed sum around each oriented face. Telescoping around a square gives $d_1 d_0 = 0$. The energy is exactly $\|d_1 f\|^2$, with no weighting or gauge penalty implicitly added.

The source also identifies this cap description with the following direct plaquette rule in $\{0, 1, 2\}^4$: retain all unit squares incident to at least one edge in E_I, and retain their edges. It obtains the same 72 faces and 112 edges, up to row orientation. Reversing a row changes its sign and not the energy. The accompanying checker independently reconstructs this rule and verifies every incidence entry in the certificate.

Source correspondence: `OneSidedCapComplex`
`embedFace_bijective; d1_d0`

Source correspondence: `OneSidedSourceRows`
`row_transport; energy_transport`

2 Exactness by paths, not an assumed gauge quotient

Lemma 2.1 (Every closed field is a gradient)**.** *For this cell, $d_1 f = 0$ if and only if $f = d_0\phi$ for some potential ϕ. The kernel of d_0 consists exactly of constant potentials. Consequently* $\operatorname{rank} d_0 = 47$ *and* $\operatorname{rank} d_1 = 65$.

Proof. For a closed field, first restrict it to the base cube. Define $\phi(v)$ by summing its edge values from the origin to v, flipping the nonzero coordinates in increasing order. Any two orders of these coordinate flips can be related by adjacent interchanges. Interchanging flips in distinct directions changes the path sum by the signed curl around their inner square, which is zero. Thus the sum is independent of their order.

For any base edge (v, i) with $v_i = 0$, a path to $v + e_i$ may therefore be chosen as a path to v followed by that edge. It follows that $\phi(v + e_i) - \phi(v) = f(v, i)$. This proves the gradient identity on every interior edge. For a cap vertex $v + e_j$, define $\phi(v + e_j) = \phi(v) + f(v, j)$, where this last edge is the outward one. This makes the gradient agree with f on all outward edges. The zero-curl equation on each ribbon then forces agreement on its tangential edge as well. These are all edges, so $f = d_0\phi$. The converse is the already established square telescoping identity.

The base cube is connected by coordinate paths, and each cap vertex is joined to its base vertex by an outward edge. Hence the full graph is connected. If $d_0\phi = 0$, potential differences vanish along

every edge, so all potential values are the same. Constants conversely have zero gradient. Rank–nullity gives $\operatorname{rank} d_0 = 48-1 = 47$. Exactness makes $\dim\ker d_1 = 47$, so $\operatorname{rank} d_1 = 112-47 = 65$. □

Source correspondence: `CubePathPoincare`
`path_stokes; cube_poincare`

Source correspondence: `OneSidedCapPoincare`
`d0_recover; closed_iff_gradient`

Source correspondence: `OneSidedGradientRanks`
`gradient_kernel_constant; gradient_rank; curl_rank`

The formal proof implements an ordered-path potential and its exact Stokes error formula. The adjacent-swap proof above reconstructs that same finite path mechanism. No cohomological exactness is assumed merely from the shape of a diagram.

3 The interior gauge and a legitimate pin

Write $A : \mathbb{R}^{E_I} \to \mathbb{R}^F$ for the restriction of d_1 to fields that vanish on the boundary, and $B : \mathbb{R}^{E_B} \to \mathbb{R}^F$ for its boundary restriction. Thus the full curvature is $Af + B\eta$. The interior energy is

$$\|Af\|^2 = \sum_{\text{inner squares}} (\operatorname{curl} f)^2 + \sum_{(v,i,j)\ \text{ribbons}} f(v,i)^2. \tag{1}$$

The second sum occurs because only the interior edge of a ribbon can contribute when the boundary field is zero.

Let $g \in \mathbb{R}^{E_I}$ equal one on the four edges leaving the origin and zero on the other interior edges. Its squared norm is four. It is a full gradient: take the potential ψ equal to -1 at the base origin and zero at every other vertex. None of the boundary edges is incident to the origin, so $d_0\psi$ equals g on the interior and zero on the boundary. In particular $Ag = 0$.

Lemma 3.1 (Exact interior kernel)**.** *The kernel of A is precisely $\mathbb{R}g$; hence* $\operatorname{rank} A = 31$.

Proof. If $Af = 0$, every square and ribbon summand in (1) is zero. An interior edge (v,i) away from the origin has $v_i = 0$ and some other coordinate $v_j = 1$. It belongs to a ribbon, which forces $f(v,i) = 0$. Only the four origin edges can remain. The inner square at the origin in directions i, j now has curl $f(0,i) - f(0,j)$ because its other two edges are away from the origin. Thus all four remaining values agree, so $f = cg$. Conversely $Ag = 0$, proving the kernel statement. Rank–nullity gives $32 - 1 = 31$. □

Source correspondence: `OneSidedBlockKernel`
`kernel_classification; energy_zero_iff`

Source correspondence: `OneSidedBlockRank`
`kernel_span; incidence_rank; gauge_is_gradient`

Choose the pole edge $e_* = (0,0)$. Pin its value to zero and let $L : \mathbb{R}^{31} \to \mathbb{R}^{72}$ be the resulting interior incidence matrix. For any unpinned f, subtract $f(e_*)g$. This does not change Af and makes the pole value zero. Thus

$$\operatorname{ran} L = \operatorname{ran} A. \tag{2}$$

If a pinned field has zero curvature, Lemma 3.1 says it is cg, and its zero pole value forces $c = 0$. Therefore L is injective and $C = L^T L$ is positive definite. It has an actual inverse: in finite dimension a positive-definite symmetric matrix is injective and hence bijective.

Source correspondence: `OneSidedPinnedInterior`
`lift_pin; pinned_range; pinned_injective; localGram_isUnit`

The unpinned matrix A^TA is singular. Only the proved pin or a separately specified generalized inverse can justify elimination. Equation (2) will allow us to check normal equations in the unpinned coordinates without ever inverting that singular matrix.

4 The boundary gradient and its physical complement

Let $G : \mathbb{R}^{48} \to \mathbb{R}^{80}$ be the restriction of d_0 to boundary edges. Its range is the boundary-gradient space, and define the physical boundary space by

$$\mathcal{P} = (\operatorname{ran} G)^{\perp} = \ker G^T.$$

This is a finite real linear definition. It is not an identification with a nonlinear gauge-invariant quantum state space.

Lemma 4.1 (The complete boundary gauge)**.** *The kernel of G is the two-dimensional space of potentials $\phi = a\mathbf{1} + c\psi$, where ψ is the origin potential defined above. Hence* $\operatorname{rank} G = 46$ *and* $\dim \mathcal{P} = 34$*.*

Proof. Both constants and ψ have zero boundary gradient. They are linearly independent, since ψ is zero at cap vertices and nonzero at the origin. Conversely suppose $G\phi = 0$. Then $d_0\phi$ has only interior components, and its full curl vanishes. Its interior restriction f satisfies $Af = 0$, so $f = cg$ by Lemma 3.1. As full edge fields, $d_0\phi = c\, d_0\psi$. Thus $d_0(\phi - c\psi) = 0$. Lemma 2.1 makes $\phi - c\psi$ constant, proving the complete kernel classification. Rank–nullity gives $48 - 2 = 46$ for G, and ordinary Euclidean orthogonal decomposition gives $\dim \mathcal{P} = 80 - 46 = 34$. □

Source correspondence: `OneSidedBoundaryGauge`
`boundary_kernel; boundary_gradient_rank; boundary_quotient_finrank`

Source correspondence: `OneSidedBoundaryCycles`
`physical_finrank; divergence_zero_iff`

Boundary gauge removal therefore costs 46 dimensions, not the single interior gauge dimension. Subtracting one dimension from both spaces would describe a different boundary problem.

5 Schur elimination and its exact kernel

For a boundary field η, define the pinned response, residual and Schur matrix by

$$\begin{aligned} y_\eta &= -(L^TL)^{-1}L^TB\eta, \\ r_\eta &= Ly_\eta + B\eta, \\ S &= B^TB - B^TL(L^TL)^{-1}L^TB. \end{aligned}$$

The normal equation is $L^Tr_\eta = 0$. In particular r_η is orthogonal to every pinned interior curvature.

Theorem 5.1 (Completed square and physical kernel)**.** *For all $y \in \mathbb{R}^{31}$ and $\eta \in \mathbb{R}^{80}$,*

$$\|Ly + B\eta\|^2 = \eta^TS\eta + \|L(y - y_\eta)\|^2, \qquad \eta^TS\eta = \|r_\eta\|^2. \tag{3}$$

The minimizing pinned y is uniquely y_η, S is symmetric positive semidefinite, and $\ker S = \operatorname{ran} G$*. Thus* $\operatorname{rank} S = 34$*, and its quadratic form is strictly positive on $\mathcal{P} \setminus \{0\}$.*

Proof. Write $Ly + B\eta = r_\eta + L(y - y_\eta)$. The cross term is zero by the normal equation. Also $B^T r_\eta = S\eta$, and

$$\eta^T S\eta = \langle B\eta, r_\eta \rangle = \langle r_\eta - Ly_\eta, r_\eta \rangle = \|r_\eta\|^2.$$

This proves (3). Injectivity of L gives uniqueness of its minimizer. The formula for S is symmetric because $L^T L$ and its inverse are symmetric, and the squared-residual identity proves nonnegativity.

If $\eta \in \operatorname{ran} G$, choose ϕ with $G\phi = \eta$. The full gradient $d_0\phi$ has zero curvature, so some interior completion has $Af + B\eta = 0$. Replace f by its pinned representative using (2). The minimum energy is zero, hence $r_\eta = 0$ and $S\eta = B^T r_\eta = 0$. Conversely, if $S\eta = 0$, then (3) gives $r_\eta = 0$. The full field comprising y_η and η is curl free and therefore a gradient by Lemma 2.1. Its boundary restriction is η, so $\eta \in \operatorname{ran} G$. This proves the kernel equality and the rank. If a physical η has zero energy, its residual is zero and it lies in $\operatorname{ran} G$ as well as its orthogonal complement; testing it against itself gives $\eta = 0$. This proves strict physical positivity. □

Source correspondence: `OneSidedSchurForm`
`completed_square; unique_minimizer; schur_posSemidef`

Source correspondence: `OneSidedSchurForm`
`kernel_schur; schur_rank; physical_positive`

These steps are classical Schur-complement reasoning, applied after a specific gauge pin has been proved legitimate. Gallier's primary exposition [3], Proposition 2.1 and Theorem 4.3, supplies the general positive-definite and singular-block context. Here the kernel and range conditions have been established for the actual incidence problem.

6 Physical cycles and the norm that compression retains

The certificate supplies an integer matrix $C_b \in \mathbb{Z}^{80\times 34}$. Its columns are actual boundary cycles. Two finite identities are crucial:

$$G^T C_b = 0, \qquad (C_b)_{c_i,j} = \mathbf{1}_{i=j} \quad (0 \le i, j < 34), \tag{4}$$

where $c_0, \dots, c_{33}$ are distinct chord-row indices, also supplied. The first identity puts every column in $\mathcal{P}$. The identity minor makes the columns linearly independent: if $C_b z = 0$, reading its chord rows gives $z = 0$. Since $\dim \mathcal{P} = 34$, these columns span all of $\mathcal{P}$. Every physical field is uniquely $\eta = C_b z$.

Source correspondence: `OneSidedBoundaryCycleData`
`cycle_balance; chord_identity`

Source correspondence: `OneSidedBoundaryCycles`
`cycle_injective; cycle_range; physical_representation`

Put

$$H = C_b^T C_b, \qquad K = C_b^T S C_b.$$

Then

$$\|\eta\|^2 = z^T H z, \qquad \eta^T S\eta = z^T K z. \tag{5}$$

In particular H is positive definite. The cycle coordinates are not orthonormal; replacing H by the identity would change the Rayleigh quotient and the claimed physical gap. We keep the generalized quotient $z^T K z/(z^T H z)$ throughout.

Source correspondence: `OneSidedSchurCompression`
`metric_quadratic; compressed_quadratic; gap_transfer`

The orthogonal split into gradient and divergence-free edge fields is classical graph Hodge theory. Jiang–Lim–Yao–Ye [4] describe this mechanism in their primary treatment of edge flows, Theorems 4.7–4.8. Their application is different; no statistical ranking conclusion or universal graph gap is imported into the present plaquette certificate.

7 A normal-equation certificate without a singular inverse

The companion data file contains integer matrices

$$X \in \mathbb{Z}^{32\times 34}, \quad R \in \mathbb{Z}^{72\times 34}, \quad Q \in \mathbb{Z}^{34\times 34}, \quad H \in \mathbb{Z}^{34\times 34}, \quad D = 193028.$$

Here A, B, C_b use the explicit source plaquette ordering in that file. The checked identities are

$$AX + DBC_b = R, \qquad A^T R = 0, \tag{6}$$

$$C_b^T C_b = H, \qquad R^T R = DQ. \tag{7}$$

All are integer equalities. Their exact arithmetic verification is described in Section 9. We first prove their mathematical implication.

Lemma 7.1 (Identifying the certificate with the original Schur form)**.** *Identities* (6)–(7) *imply* $K = Q/D$.

Proof. For any $z \in \mathbb{R}^{34}$, let $\eta = C_b z$ and $f = Xz/D$. The first equation says $Af + B\eta = Rz/D$. The second says this residual is orthogonal to $\operatorname{ran} A = \operatorname{ran} L$. The original minimizing residual r_η also lies in $B\eta + \operatorname{ran} A$ and is orthogonal to that range. The difference of these two residuals lies both in $\operatorname{ran} A$ and its orthogonal complement, so is zero. Therefore $r_\eta = Rz/D$. Taking its squared norm and using (7) gives

$$z^T K z = \|r_\eta\|^2 = \frac{z^T R^T R z}{D^2} = \frac{z^T Q z}{D}.$$

Both matrices are symmetric; equality of their quadratic forms for all z implies equality of matrices, by testing coordinate vectors and their pairwise sums. Hence $K = Q/D$. □

Source correspondence: `OneSidedIntegerResponseData response_equation; normal_equation`

Source correspondence: `OneSidedCertifiedResidual normal; compressed_eq_residual_gram; compressed_entry`

Source correspondence: `OneSidedBoundaryGap compressed_as_data; metric_as_data`

Notice that X uses all 32 unpinned coordinates. Its normal equations are valid without an inverse of $A^T A$. Uniqueness of the *residual*, not uniqueness of the unpinned response, is what identifies the certificate. Row orientation signs in the native cap model cancel in the residual Gram and normal equations; the checker independently verifies the canonical lattice row geometry.

8 The rational lower certificate and a sub-rounded witness

Define the integer matrix

$$T = 25000Q - 69639 \cdot 48257H. \tag{8}$$

The certificate supplies a positive integer s, positive integers $w_0, \dots, w_{33}$ and an integer matrix $V \in \mathbb{Z}^{34\times 34}$ such that

$$sT = V \operatorname{diag}(w_0, \dots, w_{33}) V^T. \tag{9}$$

The exact value of s has 2,386 decimal digits. All of it, every weight and every matrix entry are provided in the accompanying data; no floating-point factorization or approximate positivity check is used.

Theorem 8.1 (Certified physical boundary gap)**.** *For every $\eta \in \mathcal{P}$,*

$$\frac{69639}{100000}\|\eta\|^2 \le \eta^T S \eta.$$

If λ_* is the minimum physical Rayleigh quotient, then

$$\frac{69639}{100000} \le \lambda_* \le \frac{729067245}{1046910149} < \frac{1741}{2500} = 0.6964. \tag{10}$$

Proof. For any z, (9) gives $sz^T Tz = \sum_k w_k ((V^T z)_k)^2 \ge 0$. Division by $s > 0$ gives $z^T Tz \ge 0$. Since $D = 4 \cdot 48257$, direct substitution into (8) yields

$$K - \frac{69639}{100000} H = \frac{T}{25000D} = \frac{T}{4825700000}.$$

The denominator is positive. Equations (5) therefore prove the gap for every physical field.

For the upper comparison, the certificate's explicit vector is

$$\begin{aligned} z_* = (&2000, 0, 2000, 760, 2000, 2000, -704, -704, 9, -1935,\\ &-704, -704, -1935, 9, -1935, 9, -130, -130, -1935, -704,\\ &-704, -1935, -1935, 0, 0, -56, -56, -56, -56, -130,\\ &-56, -56, -130, -130)^T. \end{aligned}$$

The exact integer dot products are

$$z_*^T H z_* = 117410484, \qquad z_*^T Q z_* = 15782847719760.$$

Thus $\eta_* = C_b z_*$ is physical and nonzero, and its quotient is

$$\frac{\eta_*^T S \eta_*}{\|\eta_*\|^2} = \frac{15782847719760}{193028 \cdot 117410484} = \frac{729067245}{1046910149} < \frac{1741}{2500}.$$

The final comparison is exact cross multiplication of positive integers. The minimum exists: the unit sphere of the finite-dimensional physical space is compact and the quadratic form is continuous. Its lower and upper bounds follow from the proved inequality and this test field. □

Source correspondence: `OneSidedBoundaryPositiveData`
`scale_positive; weights_positive; weighted_gram_identity`

Source correspondence: `OneSidedBoundaryGap`
`corrected_eq; compressed_bound; boundary_gap`

Source correspondence: `OneSidedBoundaryGapWitness`
`integer_norm; integer_gram; rayleigh_value; below_rounded`

The witness quotient is approximately 0.69639906127; this decimal is only an orientation aid. In particular the rounded decimal 0.6964 is *not* a valid lower bound, even though the certified interval is less than 10^{-5} wide. The rational lower and upper tests have different roles. No assertion that the witness is an exact eigenvector or minimizer is made. The compactness statement introducing λ_* is a classical human rephrasing of the mapped quadratic-form bounds, not a new formal declaration.

9 What the finite arithmetic checks

This theorem is computer assisted at its finite integer-certificate stage. The geometric exactness, gauge pin, completed square, norm identification and deduction from a positive Gram certificate were proved above. The remaining finite input is fully specified by `certificate-data.json` and checked by `verify_certificate.py`. The JSON contains ordinary integer arrays and their original source hashes. Its SHA-256 digest, using UTF-8 text with LF newlines, is

`b1a7d776078753d95a18c451f4536b6cc4aeea2b12142fedaf8e66b972d01078`.

It includes the interior and boundary edge coordinates, source face coordinates, boundary endpoint indices, chord indices, A, B, C_b, X, R, H, Q, T, s, w, V and z_*. Directions and array indices are zero based.

The checker uses only Python's standard library and unbounded integer arithmetic, together with exact rational arithmetic for the last quotient. Its logical operations are the following finite computations.

1. Generate all unit squares in $\{0,1,2\}^4$ incident to one of the 32 interior edges; verify the 72-face, 112-edge, 48-vertex geometry and the 32/80 split against the listed coordinate arrays.
2. For every face, assign the four coefficients $+1, +1, -1, -1$ along its canonical boundary. Verify every entry of A, B against those rows.
3. Construct the boundary gradient matrix from its actual source and target vertices. Verify both equations in (4), including all 34 chord rows.
4. Multiply the integer arrays to verify every entry of (6) and (7).
5. Verify (8), check $s > 0$ and every $w_k > 0$, and verify all 1,156 entries of (9) by ordinary integer multiplication and addition.
6. Verify the two witness dot products and the rational comparisons in (10) by exact integer or fraction arithmetic.

Each matrix product uses the defining formula $(MN)_{ij} = \sum_k M_{ik}N_{kj}$. Thus no eigenvalue solver, numerical tolerance, symbolic positivity oracle or unproved matrix inversion is hidden in the finite certificate. These checks pass independently of the source's kernel-checked integer identities. The companion receipt records the data hash and the actual successful check; it is not a substitute for the analytic and geometric implications of those identities.

10 A local pin is not global unpinned invertibility

There is a concrete periodic counterexample to transferring a local pinned statement to simultaneous elimination of all unpinned block interiors. Let the periodic vertices be $(\mathbb{Z}/4\mathbb{Z})^4$, with positive coordinate edges. Partition them into side-two blocks by the coordinate labels $\lfloor v_i/2 \rfloor$. Call an edge interior if its two endpoints have the same four block labels. Define a periodic potential

$$\Phi(v) = \begin{cases} 1, & v_0 \in \{1,2\}, \\ 0, & v_0 \in \{0,3\}. \end{cases}$$

Let $f_i(v) = \Phi(v + e_i) - \Phi(v)$. It vanishes in directions $i \neq 0$. In direction zero, the values for $v_0 = 0, 1, 2, 3$ are respectively $1, 0, -1, 0$. The nonzero edges run from zero to one and from two to

three, so they are interior edges. The crossing edges from one to two and from three to zero have zero value. Hence f is nonzero and entirely supported on the simultaneously retained interiors.

Every plaquette curl of f is zero, because it is a gradient and the coordinate shifts commute. If A_{per} denotes the full periodic plaquette incidence restricted to all these interior edges, its nonzero interior field f_I satisfies $A_{\mathrm{per}} f_I = 0$. Consequently $A_{\mathrm{per}}^T A_{\mathrm{per}} f_I = 0$. It has no left inverse and no strictly positive unpinned coercivity constant: either assertion applied to f_I would force it to vanish or give positive norm bounded by zero.

Source correspondence: `PeriodicInteriorGaugeKernel`
`gauge_boundary_zero; witness_nonzero; incidence_witness_zero`

Source correspondence: `PeriodicInteriorGaugeKernel`
`hessian_witness_zero; hessian_no_left_inverse`

This is compatible with the local result. The one-sided pinned problem and the unrestricted periodic all-block interior matrix are different operators. The counterexample does not exclude a carefully defined global gauge quotient or a different pinned global estimate; those would require their own domain, kernel and coercivity proofs.

Periodic coarsening separates gauge, norm and curl energy

There is a further explicit calibration on this same periodic side-four torus, separate from the one-sided cell. Give all 1,024 fine directed edges and all 1,536 positively oriented plaquettes their unweighted Euclidean norms. Let D_4 be fine curl, $G_4 = D_4^T D_4$, and

$$\mathcal{F} = (\ker D_4)^{\perp}, \qquad E(f) = \|D_4 f\|^2, \qquad N(f) = \|f\|^2.$$

Removing the entire closed-field kernel includes harmonic fields, as well as gradients. On $\mathcal{F}$, E is positive definite. Write a fine vertex uniquely as $v = 2q + b$, where $q, b \in \{0,1\}^4$. The actual boundary average to the side-two coarse torus is

$$(\mathsf{A}f)(q,i) = \frac{1}{8} \sum_{b:\, b_i = 1} f(2q + b, i).$$

There are eight terms. With the stated unweighted norms its adjoint is

$$(\mathsf{A}^* c)(v,i) = \frac{1}{8} \mathbf{1}_{\{b_i = 1\}} c(q,i).$$

The identity follows by expanding both dot products and grouping edges by (q,i,b). In particular $\mathsf{A}\mathsf{A}^* = I/8$; the adjoint itself is not a right inverse.

Here is an entirely explicit witness. Put $x = v_0$, $y = v_1$, $e_x = (-1)^x$ and $s_y = (-1)^{\lfloor y/2 \rfloor}$, with fine coordinates taken modulo four. Define the fine scalar potential and its gradient by

$$\phi(v) = e_x s_y, \qquad d = \nabla_4 \phi.$$

Then $D_4 d = 0$, since forward coordinate shifts commute, but $c = \mathsf{A}d$ has components

$$c(q,0) = 2(-1)^{q_1}, \qquad c(q,i) = 0 \ (i \neq 0). \tag{11}$$

For direction zero, averaging fixes x odd and gives $d(v,0) = 2s_y$. For direction one the two values of b_0 cancel, while the other components vanish. Thus the coarse curl at $q = 0$ in the $(0,1)$ plane is 4. Raw averaging therefore does not descend even as a set map from the fine gradient quotient to the

coarse gradient quotient: $[d] = [0]$ on the fine side, whereas $[\mathsf{A}d] \neq [0]$ on the coarse side. This does not exclude corrected maps with different formulas.

Define fine fields a, p by

$$a(v,0) = s_y/8, \qquad a(v,i) = 0 \quad (i \neq 0), \tag{12}$$

$$p(v,0) = s_y(3 - e_x)/24, \qquad p(v,1) = -e_x(s_{y+1} - s_y)/24, \tag{13}$$

$$p(v,i) = 0 \quad (i = 2,3).$$

They satisfy the key identity

$$p = \mathsf{A}^* c - d/24. \tag{14}$$

This follows component by component from (11). In direction zero, $\mathsf{A}^* c = \mathbf{1}_{\{x \text{ odd}\}} s_y/4$ and $d(v,0) = -2e_x s_y$; in direction one the adjoint vanishes and the displayed component of p is $-d(v,1)/24$.

We next verify the full Gram and averaging identities, rather than assuming that either field is physical. The only nonzero curl plane is $(0,1)$, with

$$(D_4 a)_{01}(v) = \mathbf{1}_{\{y \text{ odd}\}} s_y/4, \qquad (D_4 p)_{01}(v) = \mathbf{1}_{\{x,y \text{ odd}\}} s_y/2.$$

For a field having only this curl plane, the two Gram components are

$$(G_4 f)(v,0) = (D_4 f)_{01}(v) - (D_4 f)_{01}(v - e_1), \qquad (G_4 f)(v,1) = (D_4 f)_{01}(v - e_0) - (D_4 f)_{01}(v).$$

All other components are zero. Substituting the two curl formulas, including the modulo-four wrap, proves

$$G_4 a = 2a, \qquad G_4 p = 6p - 4a, \qquad \mathsf{A}a = c/16, \qquad \mathsf{A}p = c/12. \tag{15}$$

The averaging equations follow respectively by fixing odd x and by cancelling e_x in the direction-one component. Since the range of $D_4^T D_4$ is orthogonal to $\ker D_4$, the first Gram equation puts a in $\mathcal{F}$; the second then puts $p = (G_4 p + 4a)/6$ there. This removes the whole kernel, without asserting that every closed periodic field is a gradient.

Every fine field in these formulas is constant in v_2, v_3, so the remaining two-dimensional sums have multiplicity 16. Explicitly,

$$\|c\|^2 = 64, \qquad \|a\|^2 = 4, \qquad \|p\|^2 = 16/3, \qquad \langle p, a \rangle = 4, \qquad E(p) = 16. \tag{16}$$

For clarity, p_0 has magnitude $1/12$ at even x and $1/6$ at odd x, while p_1 has magnitude $1/12$ exactly when y is odd. Thus its two component squared sums are $16(5/18)$ and $16(1/18)$, respectively. The curl formula gives $E(p) = 16 \cdot 4 \cdot (1/2)^2 = 16$. The other dot products follow from the same four-by-four table, and (11) gives the coarse norm directly. The companion checker independently enumerates all fine edges and faces with exact rational arithmetic.

Let $z = p/6 + a/3$. Equations (15) give $G_4 z = p$ and $\mathsf{A}z = 5c/144$. Consider the two actual physical lifts

$$j = 12p, \qquad h = \frac{144}{5} z.$$

Both belong to $\mathcal{F}$ and average to c. They are the unique minimizers of different objectives over the same affine set $\{f \in \mathcal{F} : \mathsf{A}f = c\}$. To prove this, write a feasible field as $j + r$, so $r \in \mathcal{F}$ and $\mathsf{A}r = 0$. The gradient d is closed and hence orthogonal to r; consequently (14) gives $\langle p, r \rangle = \langle c, \mathsf{A}r \rangle = 0$. It follows that

$$N(j + r) = N(j) + N(r).$$

Likewise $G_4 h = (144/5)p$, and any feasible $f = h + r$ has $\langle G_4 h, r\rangle = 0$. Thus

$$E(h + r) = E(h) + E(r).$$

The first residual norm vanishes only at zero; so does the second on $\mathcal{F}$. These are complete proofs of both attained, unique minima for this particular coarse field, without requiring a general coarse inverse or an unproved projection formula.

Using (16), one obtains $\langle z, p\rangle = 20/9$, $N(z) = 28/27$ and $E(z) = 20/9$. Hence

	N	E
minimum-norm physical lift j	768	2304
minimum-energy physical lift h	21504/25	9216/5

Energy decreases by 2304/5, but squared norm increases by 2304/25. The unrestricted minimum norm at coarse value c is $8\|c\|^2 = 512$, attained by $8\mathsf{A}^* c$. Indeed for $\mathsf{A}f = c$, $\langle f, \mathsf{A}^* c\rangle = \|c\|^2$ and $\|\mathsf{A}^* c\|^2 = \|c\|^2/8$, so expansion gives $N(f) = 512 + N(f - 8\mathsf{A}^* c)$. The physical correction therefore has a real norm cost; the unrestricted, physical norm and physical curl-energy minimizations are three different problems.

There is also an exact feasible descent calculation. Put

$$r_0 = 36p - 48a.$$

Then $\mathsf{A}r_0 = 0$ and $G_4 j = 72p - 48a = r_0 + 36p$. For every $r \in \mathcal{F} \cap \ker \mathsf{A}$ one has $\langle 36p, r\rangle = 0$, so r_0 is the orthogonal projection of $G_4 j$ onto the feasible direction space. It is therefore the constrained gradient of $E/2$, with this normalization. The same exact dot products give $N(r_0) = 2304$, $E(r_0) = 11520$, and

$$E(j - \alpha r_0) = 2304 - 4608\alpha + 11520\alpha^2.$$

The optimal line step is $\alpha = 1/5$, and it reaches the full affine energy minimizer exactly: $j - r_0/5 = h$. This one-step statement is for the specified witness, not a general iteration rate or scale-uniform algorithm.

Source correspondence: `PeriodicCoarseGaugeDiagnostics`
`averaged_bad_curl; no_raw_quotient_map`

Source correspondence: `PeriodicFineEnergyWitness`
`gram_a; gram_p; projectedLift_formula; average_z`

Source correspondence: `PeriodicStrictEnergyWitness`
`norm_lift_source; energy_lift_source; distinct_minimizers`

Source correspondence: `PeriodicEnergyNormTradeoff`
`energy_decrease; squared_norm_increase`

Source correspondence: `PeriodicExactDescentWitness`
`one_step_exact; optimal_step`

The printed planar formulas are a human reconstruction of the frozen integer certificates on the actual lattice. Orthogonal minimization and Hodge/Schur mechanisms are classical [3, 4]; priority for the particular witness is not certified. None of these finite linear identities concerns a nonlinear continuum gauge measure or mass gap.

11 A separate perturbation boundary: missing linear loss

Even finite positive blocks do not allow a general perturbation estimate that drops coupling cross terms. For a scalar block

$$H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad c > 0, \qquad \mathcal{S}(a,b,c) = a - b^2/c,$$

changing b to $b+d$ at $c=1$ gives the exact identity

$$\mathcal{S}(a,b+d,1) = \mathcal{S}(a,b,1) - 2bd - d^2. \tag{17}$$

At $a=2, b=1, d=1/10$, both blocks are positive definite: their Schur values are 1 and $79/100$. But

$$\frac{79}{100} < 1 - 4(1/10)^2 = \frac{96}{100}.$$

Thus a proposed loss of only $4d^2$ relative to an arbitrary nonzero reference coupling is already false in dimension two. More generally $2d + d^2 > 4d^2$ for $0 < d < 2/3$, so arbitrarily small positive perturbations have the same defect. For $|d| \le \varepsilon$, (17) instead gives the valid bound

$$\mathcal{S}(a,b+d,1) \ge \mathcal{S}(a,b,1) - 2|b|\varepsilon - \varepsilon^2.$$

Source correspondence: `ManuscriptSchurDiagnostics fixed_interior_increment; general_stability_counterexample`

Source correspondence: `ManuscriptSchurDiagnostics small_coupling_counterexample; fixed_interior_lower_bound`

For comparison, quadratic loss is justified at a zero reference coupling. If a real Hilbert-space block energy is

$$\mathcal{E}(x,y) = ax^2 + 2x\langle b, y\rangle + \langle Cy, y\rangle,$$

with $a \ge a_0 - \delta$, $\|b\| \le \varepsilon$ and $C \ge mI$, $m > 0$, then completing the scalar square yields

$$\begin{aligned} \mathcal{E}(x,y) &\ge (a_0 - \delta)x^2 - 2|x|\varepsilon\|y\| + m\|y\|^2 \\ &\ge \left(a_0 - \delta - \frac{\varepsilon^2}{m}\right) x^2. \end{aligned}$$

If instead $C_0 \ge m_0 I$ and $\|C - C_0\| \le \eta \le m_0/2$, Cauchy–Schwarz gives $\langle Cy, y\rangle \ge (m_0 - \eta)\|y\|^2 \ge (m_0/2)\|y\|^2$. The same proof gives loss $\delta + 2\varepsilon^2/m_0$. These are conditional perturbation statements, with a zero-coupling reference and proved interior coercivity. They do not automatically transport the numerical cell gap to nonlinear fields or other geometries.

Source correspondence: `CoerciveSchurStability operator_quadratic_perturbation; operator_robust_lower`

A valid comparison when the interior operator also varies

The nonzero-reference obstruction has a positive replacement. Let E be a real Hilbert space, and let $C, C_0 : E \to E$ be bounded linear operators. Assume C is symmetric and

$$\langle Cy, y\rangle \ge c\|y\|^2, \qquad \langle C_0 y, y\rangle \ge c_0\|y\|^2, \qquad c, c_0 > 0.$$

Let $a, a_0 \in \mathbb{R}$ and $b, b_0 \in E$ satisfy

$$|a - a_0| \le \delta, \quad \|b - b_0\| \le \varepsilon, \quad \|C - C_0\| \le \eta.$$

These hypotheses force $\delta, \varepsilon, \eta \ge 0$. For the moment assume the actual solved equations

$$Cu = b, \qquad C_0 u_0 = b_0, \qquad Cv = b_0.$$

We do not hide existence of these solutions in inverse notation. Put $S = a - \langle b, u\rangle$ and $S_0 = a_0 - \langle b_0, u_0\rangle$. Then

$$S \ge S_0 - \delta - \frac{\varepsilon^2 + 2\varepsilon\|b_0\|}{c} - \frac{\eta\|b_0\|^2}{cc_0}. \tag{18}$$

The proof uses coercivity directly. Cauchy–Schwarz gives $c\|w\|^2 \le \langle Cw, w\rangle \le \|Cw\|\|w\|$, and therefore $\|w\| \le \|Cw\|/c$, including $w = 0$. Consequently

$$\|u - v\| \le \varepsilon/c, \qquad \|v\| \le \|b_0\|/c, \qquad \|u_0\| \le \|b_0\|/c_0.$$

Since $C(v - u_0) = -(C - C_0)u_0$, the same argument proves

$$\|v - u_0\| \le \frac{\eta\|b_0\|}{cc_0}.$$

Writing $d = b - b_0$, symmetry of C and $C(u - v) = d$, $Cv = b_0$ give the exact identity

$$\langle b, u\rangle - \langle b_0, u_0\rangle = \langle d, u - v\rangle + 2\langle d, v\rangle + \langle b_0, v - u_0\rangle.$$

Indeed the cross terms agree because $\langle d, v\rangle = \langle C(u - v), v\rangle = \langle u - v, Cv\rangle = \langle b_0, u - v\rangle$. Bounding the three terms by the preceding norms and using $a - a_0 \ge -\delta$ proves (18).

This comparison also has its actual variational meaning. For every $x \in \mathbb{R}$ and $y \in E$, symmetry and $Cu = b$ give

$$ax^2 + 2x\langle b, y\rangle + \langle Cy, y\rangle = Sx^2 + \langle C(y + xu), y + xu\rangle. \tag{19}$$

Thus the minimum over y is attained at $y = -xu$ and equals Sx^2. Coercivity also proves uniqueness: equality forces $y + xu = 0$. No spectral-radius assertion is involved.

In finite dimension the three required equations have unique solutions: coercivity makes C and C_0 injective, and equal finite source and target dimension makes them surjective. This supplies existence rather than assuming an inverse-norm estimate. In a general Hilbert space the displayed comparison retains the supplied solutions; no finite-dimensional surjectivity argument is transferred there.

For the frequently used half-coercivity hypotheses $C_0 \ge mI$, $m > 0$, and $\|C - C_0\| \le \eta \le m/2$, one has $C \ge (m/2)I$. Taking $c = m/2, c_0 = m$ in (18) gives

$$S \ge S_0 - \delta - \frac{2\varepsilon^2 + 4\varepsilon\|b_0\|}{m} - \frac{2\eta\|b_0\|^2}{m^2}. \tag{20}$$

In finite dimension, combining this with (19) bounds the full block energy below by the right side of (20) times x^2, for every interior vector y. The direct argument above reconstructs the actual solved-equation theorem. The uniqueness statement is an elementary consequence of its printed coercive square, labelled here as a human deduction.

Source correspondence: `VaryingSchurStability`
`solved_increment; interior_solution_difference; solved_lower`

Source correspondence: `VaryingSchurStability`
`completed_square; attained_reduction; finite_solved_lower`

Source correspondence: `VaryingSchurStability`
`corridor_solutions; corridor_energy_lower`

The two linear losses have distinct sources. With $b_0 = 0$ both disappear, leaving the quadratic coupling loss, as in the sparse example below. With fixed nonzero b_0, even changing only the interior can have linear effect. The elementary scalar choice $a = a_0 = 2$, $b = b_0 = 1$, $C_0 = 1$, $C = 1 - \eta$, $0 < \eta < 1/2$, gives

$$S_0 = 1, \qquad S = 1 - \frac{\eta}{1-\eta} > 0.$$

The exact loss $\eta/(1-\eta)$ is linear at zero; it cannot be bounded by a fixed multiple of η^2 on all sufficiently small positive η. This scalar specialization is a human deduction, not an additional named source theorem. These classical Schur estimates [3] still require actual bounds for $\delta, \varepsilon, \eta, m$ in any nonlinear application. They do not supply such bounds for the full Wilson integral, another gauge geometry or varying volume.

12 An actual sparse Hessian: linear error and quadratic elimination

The preceding perturbation discussion has a direct nonlinear calibration on the same cell. Use the following normalized skew-Hermitian matrices:

$$J_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad T_j = J_j/2.$$

Thus $J_0 J_1 = J_2$, $J_1 J_0 = -J_2$, $J_j^2 = -I$ and $-2 \operatorname{Re} \operatorname{Tr}(T_j T_k) = \delta_{jk}$. Put $R_\theta = \exp(2\theta T_2)$ and $W(U) = 1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} U$. All exponentials below lie in $SU(2)$, since their exponents are traceless and skew-Hermitian.

Let $v = (1, 0, 0, 0)$ and mark three distinct interior edges

$$e_R = (v, 1), \qquad e_Y = (v + e_1, 2), \qquad e_X = (v + e_2, 1).$$

Set their links to $R_\theta, \exp(sT_1), \exp(-tT_0)$, respectively, and all other interior and boundary links to I. Inner face orientations are $i < j$ as above. A ribbon may be oriented either way, because $W(U^{-1}) = W(U)$ for $U \in SU(2)$. Deleting identity letters, the complete 72-face word inventory is

word	$\varnothing$	R	R^{-1}	X	X^{-1}	Y	Y^{-1}	RYX^{-1}
count	60	2	1	3	1	2	2	1

Here $X = \exp(-tT_0)$ and $Y = \exp(sT_1)$. This table can be obtained directly by traversing the 24 inner squares and 48 ribbons; the accompanying exact checker enumerates those traversals. It retains every face, including the 60 zero-action words. Consequently the actual full block action on this two-coordinate slice is

$$\begin{aligned} A_\theta(s,t) &= W(R_\theta e^{sT_1} e^{tT_0}) + 4(1 - \cos(s/2)) + 4(1 - \cos(t/2)) + 3(1 - \cos\theta) \\ &= 12 - 3c - c\cos(s/2)\cos(t/2) - d\sin(s/2)\sin(t/2) - 4\cos(s/2) - 4\cos(t/2), \end{aligned} \tag{21}$$

where $c = \cos\theta$ and $d = \sin\theta$. The last equality follows by multiplying the displayed matrices and using $e^{uT_j} = I\cos(u/2) + J_j \sin(u/2)$.

Source correspondence: `SparseBlockWilsonAction`
`canonical_faceWord; wordClass_card; blockAction_eq; linkMatrix_mem`

Source correspondence: `SparseBlockPartialHessian`
`blockTwo_eq; blockTwo_diag`

In particular, ordinary partial differentiation gives the symmetric Hessian at $(s,t)=(0,0)$:

$$H_\theta=\begin{pmatrix}a & b\\ b & a\end{pmatrix},\qquad a=\frac{4+\cos\theta}{4},\quad b=-\frac{\sin\theta}{4},\quad H_0=\frac{5}{4}I. \tag{22}$$

The two coordinate directions have disjoint unit edge support. Each has incidence energy five: count its incident squares/ribbons in the same face list. They are orthogonal to the origin gauge star because neither edge leaves the origin. Thus their physical squared norm is s^2+t^2, and the incidence reference energy divided by four is $\frac{5}{4}(s^2+t^2)$; no coordinate normalization is hidden here. The error $H_\theta-H_0$ has eigenvectors $(1,1),(1,-1)$ and eigenvalues

$$\frac{\cos\theta-1-\sin\theta}{4},\qquad \frac{\cos\theta-1+\sin\theta}{4}.$$

Their derivatives at zero are $-1/4$ and $1/4$. In particular the full error is not of quadratic order. Mixed partial symmetry has not removed the ordered-product contribution. Both orders differentiate the same fixed word $R_\theta e^{sT_1}e^{tT_0}$; exchanging the order of differentiation does not exchange its two matrix factors. Nevertheless $H_\theta\geq\frac{1}{2}I$ for every real θ: $a-|b|\geq(4-1-1)/4$. Linear perturbation is not loss of positivity.

Source correspondence: `SparseBlockPartialHessian`
`partialHessian_eq; error_eigen_same; error_eigen_opposite`

Source correspondence: `SparseBlockPartialHessian`
`coordinate_norm_form; coordinate_reference_form; restricted_coercive`

There is also an arbitrarily-small-field obstruction to a particular relative lower bound. Along $(s,t)=(u,u)$, write

$$h(\theta)=\frac{d^2}{du^2}A_\theta(u,u)\bigg|_{u=0}=2+\frac{\cos\theta-\sin\theta}{2}.$$

For any fixed $C\geq0$, $a_0>0$ and $\varepsilon>0$, there is $0<\theta<\varepsilon$ such that

$$h(\theta)<\frac{5}{2}\frac{\sin(a_0\theta)}{a_0\theta}-2C(a_0\theta)^2. \tag{23}$$

Here is an explicit proof, with no asymptotic remainder assumed. Let $K=5a_0^2/12+2Ca_0^2$ and choose a positive

$$\theta\leq\min\left\{\frac{\varepsilon}{2},\frac{1}{2},\frac{\pi}{12},\frac{1}{4(K+1)}\right\}.$$

For $x\geq0$, $\sin x\geq x-x^3/6$. This follows from $1-\cos x\leq x^2/2$, obtained by integrating $\sin u\leq u$, and then integrating once more. Thus the right side of (23) is at least $5/2-K\theta^2$, while $h(\theta)\leq5/2-\theta/2+\theta^3/12$. Their difference is at least $\theta(1/2-K\theta-\theta^2/12)>0$. At the base point $s=t=0$, every face holonomy is I,R_θ or R_θ^{-1}. For $|\theta|<\pi$ their principal logarithms are $0,2\theta T_2,-2\theta T_2$, respectively. In the metric $\|L\|^2=-2\operatorname{Re}\operatorname{Tr}(L^2)$, their lengths are at most $2\theta<3\theta<\pi/2$. Hence taking $a_0=3$ in (23) gives violations inside arbitrarily small strict diagonal-principal-log caps, not merely at a distant field. This tests the displayed proposed estimate; it is not a contradiction to an independently established general small-field theorem.

Source correspondence: `SparseBlockRelativeObstruction`
`arbitrary_constant_violation; no_local_relative_bound`

Source correspondence: `SU2ExplicitRotationLog`
`exp_explicitLog; explicitLog_metric_sq; diagonalLog_rotation`

Source correspondence: `SparseBlockGoodRegionObstruction`
`arbitrarily_small_strict_region`

The scalar Schur complement still has quadratic loss

Eliminate the second of these two coordinates only. Since $a \geq 3/4$, the scalar Schur complement is

$$S_\theta = a - \frac{b^2}{a} = \frac{15 + 8c + 2c^2}{4(4+c)}. \tag{24}$$

Direct factorization gives

$$S_\theta - \frac{3}{4} = \frac{(c+1)(2c+3)}{4(4+c)} \geq 0, \qquad \frac{5}{4} - S_\theta = (1-c)\frac{2c+5}{4(4+c)}.$$

For $-1 \leq c \leq 1$, the last ratio is nonnegative and at most $7/20$, by multiplication by the positive denominator. Using $1 - \cos\theta \leq \theta^2/2$ proves

$$\frac{3}{4} \leq S_\theta \leq \frac{5}{4}, \qquad \frac{5}{4} - S_\theta \leq \frac{7}{40}\theta^2. \tag{25}$$

The endpoint bounds are attained at $\theta = \pi$ and $\theta = 0$. For all real x, y, completing the square gives the exact identity

$$(x,y)H_\theta(x,y)^T = S_\theta x^2 + a\left(y + \frac{b}{a}x\right)^2.$$

Its minimum over y is $S_\theta x^2$, attained at $y = -bx/a$. This is the classical Schur mechanism [3], applied to an actual nonlinear slice. At the reference $b = 0$, its coupling enters as b^2/a, while the full Hessian retains the linear mixed term. Thus (23) and (25) are compatible. This two-coordinate elimination is distinct from the 31-coordinate incidence elimination and 34-dimensional boundary gap above. It supplies no estimate for all nonlinear link directions or for integrating out the remaining links.

Source correspondence: `SparseBlockSchurCurvature`
`schur_cos_form; schur_lower; schur_upper; schur_quadratic_loss`

Source correspondence: `SparseBlockSchurCurvature`
`actual_hessian_completed_square; actual_hessian_minimum`

13 Nonlinear Haar integration: which parity actually survives?

The quadratic result above concerns real incidence matrices. We next test two proposed nonlinear extensions on the *same* cell, using genuine $SU(2)$ link variables. Write $r(U) = \frac{1}{2}\operatorname{Tr} U$, which is real, and identify $SU(2)$ with the unit quaternions. Thus r is the real quaternion coordinate, $|r| \leq 1$, and normalized Haar measure dU is uniform measure on S^3. Put

$$a_j(t) = \cos t + \mathbf{e}_j \sin t \quad (j = 0, 1, 2), \qquad \mathbf{e}_0\mathbf{e}_1 = \mathbf{e}_2,$$

with the usual quaternion multiplication and $\mathbf{e}_j^2 = -1$. Direct multiplication gives

$$r(a_0(x)a_1(y)a_2(z)) = \cos x \cos y \cos z - \sin x \sin y \sin z. \tag{26}$$

The ordering of the three axes fixes the sign.

For each inner square use its ordered Wilson word, and for a ribbon p with interior edge e use

$$U_e b_{\mathrm{high}(p)} b_{\mathrm{outer}(p)}^{-1} b_{\mathrm{low}(p)}^{-1}.$$

Here the high and low variables are the outward cap links, and the outer variable is the tangential link. The action $\mathcal{A}_b(U)$ is the sum of $1 - r(\text{word})$ over all 72 faces. Let $\mu = \bigotimes_{e \in E_I} dU_e$, the actual product Haar law of all 32 retained links. Then

$$0 \le \mathcal{A}_b \le 144, \qquad Z_b(\beta) = \int e^{-\beta \mathcal{A}_b}\, d\mu > 0, \qquad \mathcal{F}_b(\beta) = -\log Z_b(\beta). \tag{27}$$

All real coupling derivatives can be taken under this integral: on a compact coupling interval, the kth differentiated integrand is bounded by $144^k e^{144|\beta|}$. Its power series has infinite radius for Z_b; positivity makes $\mathcal{F}_b$ real analytic at every real β. No limiting lattice or field measure enters this argument.

Source correspondence: `FourCubeHaarParity`
`fullAction_bounds; interiorHaar_probability; partition_pos`

A warning is necessary before exhibiting a cubic. If only one high cap is $a_0(x)$ and its ribbon outer link is $a_1(y)^{-1}$, all other boundary links being one, the integrated partition is separately even in x, y. Indeed conjugation by the unit quaternion $\mathbf{e}_0$ fixes $a_0(x)$ and takes $a_1(y)$ to $a_1(-y)$. Conjugate every retained link as well. Every full face word is conjugated, its trace is unchanged, and product Haar measure is preserved. Changing variables proves $Z(x, y) = Z(x, -y)$. Conjugation by $\mathbf{e}_1$ proves $Z(-x, y) = Z(x, y)$. The same identities hold for $-\log Z$, including its identically zero centered mixed difference. A pointwise cubic in a retained, *unintegrated* link cannot disprove these identities.

Source correspondence: `FourCubeHaarParity`
`turn_axis; partition_even_x; partition_even_y; freeEnergy_centered_mixed_difference`

14 A three-axis boundary response in the full 32-link integral

Use the base vertex $v = (0, 0, 1, 1)$ and its interior edge $e = (v, 0)$. Exactly two ribbons have this interior edge: $p = (v, 0, 2)$ and $p' = (v, 0, 3)$. On the boundary set the high cap of p to $a_0(x)$, the low cap of p to $a_2(z)^{-1}$, and the outer link of p to $a_1(y)^{-1}$; all other boundary links equal one. These three links are distinct. Write $\mathcal{A}_{x,y,z}$ for the resulting action and

$$W = a_0(x)a_1(y)a_2(z).$$

If $w_q = b_{\mathrm{high}(q)} b_{\mathrm{outer}(q)}^{-1} b_{\mathrm{low}(q)}^{-1}$ denotes a boundary word, this assignment gives $w_p = W$, $w_{p'} = 1$, and every other w_q is independent of y. These incidence facts can also be checked directly from $v_2 = v_3 = 1$: the allowed ribbon normal directions of e are exactly 2 and 3.

Source correspondence: `FourCubeIntegratedCubic`
`word_chosen; word_partner; word_other_y; edge_fiber`

For later use define a real open-link probe $L_{e,A}(U) = r(AU_e)$. The exact Haar identities are

$$\mathbb{E}_\mu L_{e,A} = 0, \qquad \mathrm{Cov}_\mu(L_{e,A}, L_{f,B}) = \begin{cases} \frac{1}{4} r(AB^{-1}), & e = f, \\ 0, & e \ne f. \end{cases} \tag{28}$$

To prove them, let $q \in S^3$ be the quaternion of a Haar link. Coordinate reflections give $\mathbb{E}q_i = 0$ and $\mathbb{E}q_iq_j = 0$ for $i \neq j$. Symmetry and $\sum_i q_i^2 = 1$ give $\mathbb{E}q_i^2 = 1/4$. The real coordinate of Aq is the dot product of q with $(a_0, -a_1, -a_2, -a_3)$, giving the first and same-link second identities. For distinct links independence gives zero. In particular $\operatorname{Var} L_{e,A} = 1/4$.

Let $I(U)$ be the 24-face inner action. Then

$$\mathcal{A}_b(U) = I(U) + 48 - \sum_q L_{e(q),w_q}(U), \qquad \operatorname{Cov}_\mu(I, L_{e,A}) = 0. \tag{29}$$

For the covariance claim, apply the vertex gauge transformation equal to -1 at the source of e and one at every other base vertex. The map multiplies incident link variables by central signs and preserves product Haar measure. Every closed inner face trace is unchanged by endpoint cancellation, whereas $L_{e,A}$ changes sign. Hence $\mathbb{E}[IL_{e,A}] = 0$; its covariance is zero because the probe is centered. All functions are bounded, so each change of variables and covariance expansion is legitimate.

Source correspondence: `FourCubeHaarChannels`
`linkProbe_pair; linkProbe_covariance; internal_link_orthogonal; internal_link_covariance`

Theorem 14.1 (Integrated cubic response)**.** *For the actual product Haar law of all 32 links,*

$$\operatorname{Var}_\mu(\mathcal{A}_{x,y,z}) - \operatorname{Var}_\mu(\mathcal{A}_{x,-y,z}) = -\sin x \sin y \sin z, \tag{30}$$

$$\mathcal{F}''_{x,y,z}(0) - \mathcal{F}''_{x,-y,z}(0) = \sin x \sin y \sin z. \tag{31}$$

Whenever the sine product is nonzero and $\varepsilon > 0$, there is $0 < \beta < \varepsilon$ for which $Z_{x,y,z}(\beta) \neq Z_{x,-y,z}(\beta)$. Moreover the ordered boundary derivative, taking y first, is

$$\partial_z\partial_x\partial_y\mathcal{F}''_{x,y,z}(0)\Big|_{x=y=z=0} = \tfrac{1}{2}.$$

Proof. Remove just the chosen probe by defining

$$R_{x,z} = \mathcal{A}_{x,0,z} + L_{e,a_0(x)a_2(z)}.$$

Equation (29) shows that this residual is independent of y and that $\mathcal{A}_{x,y,z} = R_{x,z} - L_{e,W}$. For any unit quaternion A, expand its covariance with the residual. The inner-action term vanishes by (29); constants vanish; all ribbon probes on other edges vanish by (28). Among the probes on e, the chosen one has been removed and only its partner of word one remains, with a minus sign. Consequently

$$\operatorname{Cov}_\mu(R_{x,z}, L_{e,A}) = -\tfrac{1}{4}r(A), \qquad \operatorname{Var}_\mu \mathcal{A}_{x,y,z} = \operatorname{Var}_\mu R_{x,z} + \tfrac{1}{4} + \tfrac{1}{2}r(W).$$

Subtracting the expression with y negated and using (26) proves (30). The constant internal variance was never discarded or approximated; it cancels exactly.

Differentiating (27) gives $Z(0) = 1$, $Z'(0) = -\mathbb{E}\mathcal{A}$ and $Z''(0) = \mathbb{E}\mathcal{A}^2$. Thus $(-\log Z)''(0) = -\operatorname{Var}\mathcal{A}$, proving (31). The variance formula also gives

$$\partial_y\mathcal{F}''_{x,y,z}(0)\Big|_{y=0} = \tfrac{1}{2}\sin x \sin z.$$

Differentiating this displayed identity in x and then z at zero gives $1/2$. This order of derivatives is explicit; no interchange of boundary derivatives is required.

Finally suppose the two partitions agreed for every $0 < \beta < \varepsilon$. Their logarithms would agree there and at zero. Their twice differentiable extensions to an open neighborhood of zero would then have equal second derivatives at zero, contrary to (31). This proves the claimed arbitrarily small positive witnesses. □

Source correspondence: `FourCubeIntegratedCubic`
`variance_formula; variance_odd_y; variance_explicit_odd_witness`

Source correspondence: `FourCubeIntegratedResponse`
`freeEnergy3_second_zero; integrated_mixed_cubic`

Source correspondence: `FourCubeIntegratedResponse`
`exists_positive_coupling_nonparity_of_sine`

For $x = y = z = \pi/6$ the variance difference is $-1/8$ and the second free-energy response difference is $1/8$. Both boundary angles and positive coupling can be made arbitrarily small: choose $0 < t < \min(\delta, \pi)$ and use $x = y = z = t$, whose sine product is positive. This is a small-coupling result; it supplies no large-coupling estimate.

The underlying trace is gauge invariant in a precise sense. Under any of the 48 vertex gauges g, a boundary word along a path from the target of e to its source transforms as

$$w_q \longmapsto g(\text{target } e) w_q g(\text{source } e)^{-1}.$$

For p, p' with the same interior edge, $w_p w_{p'}^{-1}$ is therefore conjugated at the target. Its half trace, equal here to $r(W)$, is unchanged. In particular its difference $-2 \sin x \sin y \sin z$ under the tested reflection cannot be dismissed as an open-link trace which should vanish by a vertex gauge change. An open probe alone is not gauge invariant; its same-edge covariance produces this closed path.

Source correspondence: `FourCubeIntegratedCubic`
`word_boundaryGauge; pair_trace_gauge; pair_trace_odd_y`

15 An exact conditional calculation at large coupling

Now keep the same three-axis boundary, but fix the other 31 retained links to one and integrate only $U_e = U$. Denote the resulting partition and free energy by $\widehat{Z}$ and $\widehat{\mathcal{F}}$. They are conditional kernels evaluated at this configuration, defined pointwise by a finite integral. Conditioning at this particular Haar-null configuration does not identify them with the full marginal of Theorem 14.1.

The chosen edge is incident to three inner squares, each contributing $1 - r(U)$. Exactly two ribbons use that edge; their words are W and one. Among the remaining ribbons, two have half trace $\cos x$, two have half trace $\cos z$, and the remaining 42 have trace one. It follows, retaining all 72 faces, that

$$\mathcal{A}_{x,y,z}(U, 1, \dots, 1) = c(x, z) - 4r(U) - r(UW), \quad c(x, z) = 5 + 2(1 - \cos x) + 2(1 - \cos z). \tag{32}$$

For clarity, at $U = 1$ the 48 ribbon traces are classified as $r(W)$ once, $\cos x$ twice, $\cos z$ twice and one 43 times. The partner ribbon lies in the last class; releasing U changes it from trace one to $r(U)$ and changes the chosen trace to $r(UW)$. This also verifies (32) without subtracting any boundary-dependent constant.

Source correspondence: `FourCubeConditionalAction`
`internal_single; capClass_count; ribbonSum_one; ribbonSum_change; conditional_action`

Define the one-link moment generating function and tilted mean

$$M(t) = \int e^{tr(V)}\, dV, \qquad m(t) = \frac{M'(t)}{M(t)}, \qquad \rho(x, y, z) = \sqrt{17 + 8r(W)}.$$

The quaternion $4+W$ has squared norm $17+8r(W)$, so $3 \le \rho \le 5$. Its unit orientation $n = (4+W)/\rho$ gives $4r(U) + r(UW) = \rho r(Un)$. Haar translation in (32) proves the exact polar reduction

$$\widehat{Z} = e^{-\beta c(x,z)} M(\beta\rho), \qquad \widehat{\mathcal{F}} = \beta c(x, z) - \log M(\beta\rho). \tag{33}$$

$M > 0$ and all its real derivatives exist, since $|r(V)| \le 1$. Also $m(t)$ is the expectation of r under the probability density $e^{tr}/M(t)$, hence $|m(t)| \le 1$.

Source correspondence: `FourCubeConditionalRadial`
`stapleRadius_sq; stapleRadius_bounds; five_trace; radial_integral`

Source correspondence: `FourCubeConditionalResponse`
`conditionalPartition_formula; conditionalFreeEnergy_formula`

Lemma 15.1 (Concentration of the one-link mean)**.** *$m(t) \longrightarrow 1$ as $t \longrightarrow +\infty$.*

Proof. For $0 < \delta < 1$, the cap $C_\delta = \{r > 1 - \delta/2\}$ is a nonempty open neighborhood of the identity. Haar measure has full support: if a nonempty open set had measure zero, finitely many of its left translates would cover the compact group, contradicting total mass one. Thus $c_\delta = \mu_H(C_\delta) > 0$. For $t \ge 0$,

$$M(t) \ge c_\delta e^{t(1-\delta/2)}, \qquad \int_{\{r \le 1-\delta\}} e^{tr}\, d\mu_H \le e^{t(1-\delta)}.$$

The tilted mass of $\{r \le 1 - \delta\}$ is at most $c_\delta^{-1} e^{-t\delta/2}$. On its complement $1 - r < \delta$, and everywhere $0 \le 1 - r \le 2$. Therefore $0 \le 1 - m(t) \le \delta + 2c_\delta^{-1} e^{-t\delta/2}$. First take $t \to \infty$, then $\delta \downarrow 0$. □

Source correspondence: `SU2OneWordLargeCoupling`
`positive_mean_tendsto_one`

Let the ordered mixed conditional Hessian be

$$H_\beta(z) = \partial_x \partial_y \widehat{\mathcal{F}}(\beta, x, y, z)\Big|_{x=y=0}, \qquad \rho_z = \sqrt{17 + 8\cos z}.$$

Differentiating the positive square root in (26) gives $\partial_y \rho(x, 0, z) = -4 \sin x \sin z / \rho(x, 0, z)$. The constant c has no y dependence. By (33),

$$\partial_y \widehat{\mathcal{F}}\Big|_{y=0} = \frac{4\beta m(\beta\rho(x, 0, z))}{\rho(x, 0, z)} \sin x \sin z.$$

The coefficient is differentiable in x because $\rho \ge 3$ and m is differentiable. At $x = 0$ its derivative is multiplied by $\sin 0$, so the exact result is

$$H_\beta(z) = \frac{4\beta m(\beta\rho_z)}{\rho_z} \sin z, \quad H_\beta(0) = 0, \quad \frac{H_\beta(z)}{\beta} \longrightarrow \frac{4 \sin z}{\rho_z}. \tag{34}$$

Similarly the ordered third boundary derivative at zero is $H'_\beta(0) = \frac{4}{5}\beta m(5\beta)$, whose ratio to β tends to $4/5$. These statements concern the conditional integral only.

Source correspondence: `FourCubeConditionalResponse`
`conditional_mixed_hessian; conditional_mixed_cubic`

Source correspondence: `FourCubeConditionalResponse`
`conditional_mixed_cubic_normalized_limit`

16 A failed quadratic budget and the valid linear one

Theorem 16.1 (Conditional boundary budget obstruction)**.** *There do not exist $C \ge 0$, $C_0 \in \mathbb{R}$, $\varepsilon > 0$ and $\beta_0 \in \mathbb{R}$ such that*

$$|H_\beta(z)| \le C\beta z^2 + C_0 \quad \textit{for all } 0 < z < \varepsilon, \quad \beta > \max(0, \beta_0).$$

The obstruction persists on $\beta = s^2$, $z = 1/s$: for all sufficiently large s, $H_{s^2}(1/s) \geq s/5$. Nevertheless the global bound

$$|H_\beta(z)| \leq \tfrac{4}{3}|\beta|\,|z| \qquad (\beta, z \in \mathbb{R})$$

holds.

Proof. Choose a positive $z < \min(\varepsilon, 1)$ so small that $Cz^2 < 2z/5$. Such a choice exists for every $C \geq 0$. The elementary inequality $\sin z \geq z - z^3/6 \geq z/2$ and $\rho_z \leq 5$ give $4 \sin z/\rho_z \geq 2z/5 > Cz^2$. The limit in (34) then implies $H_\beta(z) > C\beta z^2 + C_0$ for all sufficiently large positive β: after dividing by β, the constant C_0/β vanishes. This contradicts the proposed absolute-value bound.

For the shrinking corridor, $s^2\rho_{1/s} \geq 3s^2 \to \infty$, so eventually $m(s^2\rho_{1/s}) \geq 1/2$. For $s \geq 2$, $\sin(1/s) \geq 1/(2s)$. Inserting these bounds and $\rho_{1/s} \leq 5$ in (34) gives

$$H_{s^2}(1/s) \geq \frac{4s^2}{5}\,\frac{1}{2}\,\frac{1}{2s} = \frac{s}{5}.$$

On this corridor $C\beta z^2 + C_0 = C + C_0$, which any sufficiently large $s/5$ exceeds, even for arbitrary real C, C_0. Finally $|m| \leq 1$, $\rho_z \geq 3$ and $|\sin z| \leq |z|$ in (34) prove the stated linear bound. □

Source correspondence: `FourCubeConditionalObstruction no_uniform_quadratic_budget; eventually_shrinking_lower`

Source correspondence: `FourCubeConditionalObstruction exists_shrinking_budget_violation; mixedHessian_linear_bound`

These results separate exact two-axis parity, a full-integral cubic response near zero coupling, and a conditional large-coupling budget obstruction with the other 31 links fixed. Their ensembles remain distinct. The failure of a βz^2 budget does not disprove a Schur-complement estimate with an allowed linear remainder, and does not invalidate the rational finite quadratic gap proved earlier.

The methods themselves have classical antecedents. Creutz's primary paper [1], Section II, equations (2.2)–(2.7), specifies the product group integral, Wilson plaquette action and one-link conditional update. Equations (2.12)–(2.18) explicitly reduce a sum of $SU(2)$ staples to a radial amplitude and a unit orientation. Our $4 + W$ is a five-staple instance of that mechanism; the finite cap cell has a different incidence from the periodic six-staple bulk example. Creutz [2], page 2043, equations (1)–(4), also gives the classical invariant-integration and generating-function framework. We use elementary S^3 moments and bounded differentiation here, not a new integration algorithm. The exact boundary witnesses and their proposed-estimate diagnostics are traced to the frozen source; their formulation priority remains unresolved. None of these finite results supplies scale or volume uniformity, a continuum construction, or a physical spectral mass gap.

17 Exact Haar pinning and full-cell vacuum concentration

The linear pin used above has a genuine nonlinear counterpart. Let $e_* = (0, 0)$ be the pole edge, let $E_* = E_I \setminus \{e_*\}$, and put $\mathcal{Y} = SU(2)^{E_*}$, with product normalized Haar probability ν. Thus $|E_*| = 31$. Write $\mathrm{fill}(V, y)$ for the interior field whose pole is V and whose other links are y. For fixed boundary links b, denote the full 72-face action by $\mathcal{A}_b$.

A gauge equal to V at the origin and identity at every other vertex left multiplies all four outgoing origin links by V. It fixes all other links, including boundary links. Each face word is conjugated at its base vertex, so its real trace is unchanged. On the 31 free links let R_V left multiply precisely the other three origin links. Then

$$\mathcal{A}_b(\mathrm{fill}(V, R_V y)) = \mathcal{A}_b(\mathrm{fill}(I, y)).$$

The map R_V preserves ν by coordinatewise Haar invariance. Split the 32-link product integral into its pole and free coordinates. Changing variables by R_V in each free-coordinate integral gives

$$\int_{SU(2)^{32}} e^{-\beta \mathcal{A}_b(U)} dU = \int_{\mathcal{Y}} e^{-\beta \mathcal{A}_b(\mathrm{fill}(I,y))} d\nu(y). \tag{35}$$

This holds for every real β and every fixed boundary b. Indeed the action is continuous between zero and 144, so the integrands are positive, bounded and integrable; Fubini and the changes of variables are justified. The integrated pole has Haar mass one. This equality of partitions does not identify untransformed coordinate observables in the two ensembles.

Source correspondence: `FourCubePinnedHaar`
`rootLeft_preserving; fullAction_fillPole; conditional_pole_integral`

Source correspondence: `FourCubePinnedHaar`
`partition_eq_pinned; pinnedPartition_positive`

For concentration we now fix *identity boundary*. Define

$$S(y) = \mathcal{A}_1(\mathrm{fill}(I,y)), \quad d(U) = 1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U, \quad Z_\beta = \int e^{-\beta S} d\nu, \quad d\mu_\beta = Z_\beta^{-1} e^{-\beta S} d\nu.$$

The action is the sum of the 24 inner-face defects and 48 ribbon link defects, with multiplicities retained. It is nonnegative, continuous, at most 144 and zero at the identity configuration. The normalizer is strictly positive, so μ_β is an actual probability law, not an unnormalized Boltzmann expression.

Lemma 17.1. *For every free link e and configuration y,*

$$d(y_e) \le 3S(y), \qquad S(y) = 0 \iff y = I.$$

Proof. Under the unit-quaternion model $SU(2) \cong S^3$, $\|U - I\|^2 = 2d(U)$, so $d \ge 0$ and $d(U) = 0$ exactly at I. Multiplicativity of quaternion norm and the telescoping identity for $ABC - I$ give

$$\|ABC - I\| \le \|A - I\| + \|B - I\| + \|C - I\|.$$

Squaring and using $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ gives $d(ABC) \le 3(d(A) + d(B) + d(C))$. Also $d(U^{-1}) = d(U)$.

Every interior edge away from the origin belongs to a ribbon: its base has a nonzero coordinate other than its own edge direction. That ribbon contributes precisely its link defect to S, so $d(y_e) \le S$ for these links. For a remaining origin link y_c in direction $j \ne 0$, use the inner square based at the origin in directions $0, j$. The pole link is identity, so its face word is

$$Q = y_a y_b^{-1} y_c^{-1}, \qquad a = (e_0, j), \quad b = (e_j, 0), \quad c = (0, j).$$

Hence $y_c = Q^{-1} y_a y_b^{-1}$. The two distinct shifted edges a, b each occur in a ribbon; choose one such ribbon for each. They are distinct because their interior edges are distinct. The sum of their defects is at most the total ribbon energy, and $d(Q)$ is at most the total inner-face energy. The triple-product inequality therefore gives $d(y_c) \le 3S$. This covers every free edge. If $S = 0$, every free-link defect is zero, hence every free link is identity. The converse follows by substituting identity into every word. □

Source correspondence: `FourCubePinnedCoercivity`
`defect_zero_iff; defect_triple; off_origin_bound`

Source correspondence: `FourCubePinnedCoercivity`
`pinned_coordinate_bound; pinnedAction_zero_iff; global_energy_barrier`

Theorem 17.2. *Let $\epsilon > 0$, and set*

$$m_\epsilon = \nu(\{y : S(y) < \epsilon/6\}) > 0.$$

For every $\beta \geq 0$,

$$\mu_\beta\{y : \max_{e \in E_*} d(y_e) \geq \epsilon\} \leq \frac{31}{m_\epsilon} e^{-\beta\epsilon/6}. \tag{36}$$

Thus all 31 pinned links simultaneously approach identity in probability as $\beta \to \infty$. Moreover μ_β converges weakly to the identity point mass on the fixed compact space $\mathcal{Y}$; this last implication is a human deduction.

Proof. The strict sublevel set in m_ϵ is a nonempty open set, since S is continuous and $S(I) = 0$. Product Haar measure gives positive mass to every such set. For completeness, a nonempty open set in a compact group contains a translate of a nonempty identity neighborhood. Finitely many translates of that neighborhood cover the group. If its Haar mass were zero, invariance and this finite cover would give total mass zero. The argument applies to the finite product group, proving $m_\epsilon > 0$.

On that sublevel set $e^{-\beta S} \geq e^{-\beta\epsilon/6}$, so

$$Z_\beta \geq m_\epsilon e^{-\beta\epsilon/6}.$$

On a fixed link tail $d(y_e) \geq \epsilon$, Lemma 17.1 gives $S \geq \epsilon/3$. Its unnormalized integral is therefore at most $e^{-\beta\epsilon/3}\nu(\{d(y_e) \geq \epsilon\}) \leq e^{-\beta\epsilon/3}$. Dividing by the proved lower normalizer bound gives $\mu_\beta\{d(y_e) \geq \epsilon\} \leq m_\epsilon^{-1} e^{-\beta\epsilon/6}$. The union bound over 31 links proves (36). For every fixed ϵ the right side tends to zero.

Finally $d(U) = \frac{1}{2}\|U - I\|^2$ shows that for every neighborhood O of the identity configuration, some positive ϵ has $\{\max_e d(y_e) < \epsilon\} \subset O$. For continuous f on the compact space $\mathcal{Y}$ and $\eta > 0$, choose O so that $|f(y) - f(I)| < \eta$ on O. Then

$$\left| \int f \, d\mu_\beta - f(I) \right| \leq \eta + 2\|f\|_\infty \mu_\beta(O^c).$$

The tail estimate makes the second term vanish; letting η decrease to zero proves weak convergence. □

Source correspondence: `FourCubePinnedConcentration`
`lowMass_positive; normalizer_lower; gibbs_coordinate_tail_bound`

Source correspondence: `FourCubePinnedConcentration`
`gibbs_configuration_tail_bound; gibbs_configuration_good_tendsto_one`

The same pin also identifies the full local Hessian, without a Gaussian approximation to the integral. In the generators T_a used above let $A(v) = \sum_{a=0}^{2} v_a T_a$, so $-2\,\mathrm{Re}\,\mathrm{Tr}(A(v)A(w)) = v \cdot w$. For a free-link velocity $V = (V_e) \in (\mathbb{R}^3)^{31}$ set $y_e(t) = \exp(tA(V_e))$, extending V by zero on the pole and boundary. The finite matrix exponentials and products are C^2. For each oriented face, the first-order Lie-algebra sum is $A(\operatorname{curl} V)$. Expanding the four exponentials through second order and using $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ gives its defect $t^2|\operatorname{curl} V|^2/8 + o(t^2)$. Indeed the real trace of the quadratic coefficient is $\frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}((\sum A_i)^2)$; the linear trace is zero. Each ribbon similarly contributes $t^2|V_e|^2/8 + o(t^2)$. Summing all 72 terms, and using (1) and $C = L^T L$, proves

$$\frac{d^2}{dt^2} S(y(t))\bigg|_{t=0} = \frac{1}{4} \sum_{a=0}^{2} (V^a)^T C V^a > 0 \quad (V \neq 0).$$

The same expression is the diagonal of the actual Fréchet Hessian of $V \mapsto S((\exp A(V_e))_e)$ at zero: composition with the line tV computes that diagonal for this C^2 map. Positivity uses the already proved positive-definite pinned incidence Gram matrix. This local Hessian and the global concentration estimate are different statements; neither supplies a uniform Gaussian approximation, a partition asymptotic or a closed evolution generator.

Source correspondence: `FourCubePinnedSecondVariation`
`pinned_first_zero; pinned_second_gram; pinned_second_positive`

Source correspondence: `FourCubePinnedHessian`
`coordinateAction_contDiff; hessian_diagonal_gram; hessian_positive`

All constants concern this fixed cap cell and identity boundary. In particular the sublevel mass m_ϵ has not been bounded uniformly over volumes, boundary data or lattice scales. Compact Gibbs concentration and Haar gauge changes are classical mechanisms; the finite geometry and factors are reconstructed here without a priority claim. This does not identify the different conditional large-coupling ensemble above, or establish a physical spectral gap.

18 Provenance and remaining uniformity questions

The frozen source is commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The theorem map records exact declarations, original source hashes and human reorganizations. The original full-build and axiom-audit evidence hashes were checked; publication work changes no mathematical source and runs no competing Lean research build.

Four independent standard-library checkers retain 66 exact groups: nine certificate checks rebuild the geometry, all integer Gram/residual identities and the witness quotient; 21 Haar checks enumerate actual ribbons and rational quaternion actions; 20 sparse checks test all face words, second jets and Schur identities; and 16 periodic checks traverse the fine lattice and verify the two attained lift costs and feasible step. Their integers and fractions involve no floating arithmetic. The checker files and exact certificate data accompany the paper. Differentiation, Haar integrals, normalizer lower bounds and asymptotic claims rely on the printed proofs, not sampled integration.

All results retain their stated ensembles, metrics and boundary conditions. Sublevel masses have no claimed uniform control over volume or boundary data. No varying lattice scale, continuum construction or physical spectral identification is supplied. The general mechanisms are classical; priority for the exact finite geometry and witnesses remains unresolved.

References

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